

Lab 3 — Architecture Selection & Justification

Academic Elective Bidding & Allocation System

Architecture Selection: “We chose Layered Architecture for the Academic Elective Bidding & Allocation System.”

Architectural Choice

We selected **Layered Architecture**, organised into Presentation, Business Logic, Data, and External Systems. The Web Portal provides the student and Academic Registrar interfaces; the Business Logic layer contains Authentication & RBAC, Bidding Management, Validation, Allocation Solver, and Reporting & Notification; and the Data layer provides centralised academic persistence.

Reason 1 — Clear Separation of Academic Responsibilities

The system has distinct responsibilities such as bid submission, prerequisite and timetable validation, allocation, reporting, and authentication. Layering separates these responsibilities into focused components, so changes to bidding or allocation rules can be made without directly changing the presentation interface or the underlying database structure.

Reason 2 — Suitable for the Centralised Bidding Workflow

The Academic Registrar configures the bidding window, runs the allocation solver, and views reports, while students browse electives, submit or modify bids, and view results through the Web Portal. A layered structure provides a consistent business-logic boundary for enforcing the 100-credit rule, eligibility checks, timetable constraints, allocation rules, and result generation.

Security Advantage

Authentication and role-based access control are handled by the dedicated **Authentication & RBAC Component** in the Business Logic layer. It integrates with the University SSO System and can enforce different permissions for Students and Academic Registrars before protected operations such as bid modification, configuration, and allocation are performed.

Performance Benefit

The **Allocation Solver Component** is isolated within the Business Logic layer, allowing the allocation algorithm and its data access to be implemented and optimised independently of the Web Portal. This supports efficient batch processing of student bids and makes the solver easier to benchmark against the system's required allocation-performance target.